Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – June 19th, 2023

Polimi ID _____ _____

Surname_____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Is it possible to eliminate a flow dependency by modifying the program and ensuring that the program remains equivalent? (2)

B. Is the following code parallelizable? Are there better parallelizable solutions? (2)

```
for (i=0; i<100; i++)
        for (j=1; j<100; j++)
            a[i][j] = f(a[i][j-1]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the reduce pattern: assumptions, complexity, and example. (2)

B. Please describe the differences between Gather and scatter parallel patterns. (1)

C. What is meant by "*merge scatter*"? (1)

Exercise n. 3

Answer the following questions about OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe what is meant by nested parallelism and how could be implemented in OpenMP?(2)

B. What is the difference between static and dynamic schedule clauses of OpenMP? In which OpenMP directive is it used? (2)

Exercise n. 4

Answer the following questions about MPI and PThreads and briefly explain (without explanation the answer will be considered invalid)

A. Describe the procedure MPI_Send (You can skip this question if you completed the second challenge) (2)

B. What is it meant for the procedure pthread_barrier_wait? (2)